

Boolean conditions

Project: Python readiness portfolio

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Translate stated requirements into comparisons

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate numeric expressions using Python's grouping rules

Reference example, not the only acceptable wording: $3 * 8 + 2$ or $(3 * 8) + 2$ gives 26. $3 * (8 + 2)$ gives 30 and adds two to each of three packs, not two once.

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: `packs = 4; per_pack = 6; total = packs * per_pack; print(total)`, each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24. Assignment alone produces no displayed output in this script.

Use fictional activities and Boolean operands throughout. Do not equate suitability with permission or claim that a programmed condition learns from data. Reference for teacher precision: <https://docs.python.org/3/library/stdtypes.html#boolean-operations-and-or-not>. Python and/or may return operands of other types; this lesson deliberately confines them to Booleans.

Outcomes

Translate stated requirements into comparisons

Combine Boolean values with and or and not

Choose a case that distinguishes two conditions

Preparation

Python 3 editor and conditions.py

True and False cards and written activity requirements

Blank input, separate checks, expected and observed columns

Prerequisite answer

False is a Boolean value; "False" is text. This week uses actual Boolean values for yes-or-no flags.

End goal for the pictorial case

Trace boolean conditions and explain the changed program before running it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

A fictional activity must take at most 12 minutes and be quiet. An activity taking exactly 12 minutes meets the time requirement. A quiet 15-minute activity does not. Before writing code, underline each requirement and decide whether all requirements must hold or whether alternatives are allowed.

Use a fresh `conditions.py` file with nonnegative integer minutes and an actual Boolean quiet flag. A comparison produces True or False for these values. The expression `minutes <= 12` includes 12; `minutes < 12` excludes it. Use `==` for equality and `!=` for inequality. A single `=` assigns a value to a name; it does not ask whether two values are equal.

The script prints True then False. The first result concerns time only. The second requires `time_ok` and quiet. These are separate claims. Trace each input before explaining the combined result; a result without its conditions can be misleading.

Term · Meaning

Term: Comparison

Meaning: An expression that checks a relation between values

Term: Boolean

Meaning: A type with the values True and False

Term: Conjunction

Meaning: A requirement that both conditions hold

Term: Inclusive or

Meaning: A condition satisfied when either or both parts hold

Term: Counterexample

Meaning: A case that shows a claim or rule is wrong

Two requirements must both hold

Input case · Time check · Combined result

Input case: 12 min, quiet False

Time check: 12 ≤ 12 is True

Combined result: True and False is False

Input case: 12 min, quiet True

Time check: 12 ≤ 12 is True

Combined result: True and True is True

Input case: 15 min, quiet True

Time check: 15 ≤ 12 is False

Combined result: False and True is False

Passing only one requirement does not make the activity suitable.

Inclusive or accepts both available; exactly one would reject both. Exit: (12, True) should pass and reveals ≤ changed to <. (8, False) should fail and reveals and changed to or. Other inputs are acceptable if they distinguish the versions.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

False · False · False · False

False: False

False: True

False: False

False: True

False: True

False: False

False: False

False: True

False: True

False: True

False: True

False: True

With Boolean operands, and needs both parts True. Inclusive or needs at least one True, so both True is allowed. not reverses a Boolean: not True is False and not False is True. Python can also apply and and or to other types; keep operands Boolean here so this table directly describes the result.

A game requires a pencil or a crayon. If both are available, does inclusive or accept the case? Explain how this differs from requiring exactly one.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Boolean conditions

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

The facts we will use

Python 3: `age=15; allowed=(age>=12) and (age<=15); print(allowed)`. No type validation is implemented here.

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Boolean conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check lower boundary

$15 \geq 12$ is True.



2. Check upper boundary

$15 \leq 15$ is also True.



3. Combine with and

Both comparisons are True, so allowed is True.

CHANGE ONE THING

Change age to 16.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Check lower boundary

What happens and why: $15 \geq 12$ is True.

Step: 2. Check upper boundary

What happens and why: $15 \leq 15$ is also True.

Step: 3. Combine with and

What happens and why: Both comparisons are True, so allowed is True.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A Boolean condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

What this example does not establish

This condition checks a declared numeric interval, not identity or authorisation; unexpected input types need separate handling.

Change one thing in the pictured case

Change: Change age to 16.

Prediction: The result becomes False.

Reason: The lower condition remains True, but the upper condition fails.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Check Boolean versus text with the prerequisite.

Time or stage: 7 to 20 minutes

Teacher action: Underline the two requirements and complete W1.

Time or stage: 20 to 33 minutes

Teacher action: Model the diagram and trace the two printed values.

Time or stage: 33 to 45 minutes

Teacher action: Complete W2 with predictions before running.

Time or stage: Session 2 start to 8 minutes

Teacher action: Retrieve an inclusive boundary and a one-condition failure.

Time or stage: 8 to 20 minutes

Teacher action: Build the four-row truth table and complete W3 and the guided question.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including a counterexample.

Time or stage: 35 to 45 minutes

Teacher action: Discuss W5 and use the exit to diagnose boundary and logic errors.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Use separate cards for each comparison, then combine them. Read at most and at least aloud with number lines. Extension: explain why not ((minutes $\geq$ 5) and (minutes $\leq$ 9)) identifies out-of-range numeric values under the stated input contract.

Reduce help deliberately

Completion task: For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time comparison and final suitability values before running the program.

Teacher check: Time results are True, True, False. Combined results are True, False, False; the second

case fails quiet and the third fails time.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Boolean conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: At most 12 includes 12; compare $<$ and $<=$ at exactly 12.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Give both comparison results for ages 11, 15 and 16. Which comparison rejects each outside value, and why does and matter?

Reasoned teacher answer

Age 11 gives False/True, so False. Age 15 gives True/True, so True. Age 16 gives True/False, so False. Both interval boundaries must hold.

Changed-case reasoning: The result becomes False. The lower condition remains True, but the upper condition fails.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

At most excludes the boundary: At most 12 includes 12; compare $<$ and $<=$ at exactly 12.

or means exactly one: Inclusive or also accepts two True values.

A single equals sign tests equality: $=$ assigns; $==$ compares.

A True result proves a fact about the world: It establishes only that the supplied values satisfy this expression.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Inclusive or accepts both available; exactly one would reject both. Exit: (12, True) should pass and reveals $\leq$ changed to $<$. (8, False) should fail and reveals and changed to or. Other inputs are acceptable if they distinguish the versions.

W1 Translate the requirement

Write a condition for an activity taking at most 12 minutes and being quiet. Explain why exactly 12 minutes

is included.

(minutes <= 12) and quiet. At most includes equality, so <= is required; the nonnegative integer and Boolean input contract still applies.

W2 Trace separate checks

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time comparison and final suitability values before running the program.

Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

W3 Distinguish alternatives

Complete all four Boolean combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

or gives False, True, True, True for FF, FT, TF, TT. Both available is accepted. not pencil is True when pencil is False and False when pencil is True.

W4 Write a new rule independently

A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False). Choose a case that detects replacing and with or between the range check and available.

(minutes >= 5) and (minutes <= 9) and available. Results True, True, False, False. (7, False) distinguishes (range_check and available) from (range_check or available): correct False versus faulty True.

W5 Separate a check from a guarantee

Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

The expression checks supplied values against stated conditions. Inaccurate input can still pass. Permission is a separate human decision and must not be inferred from suitability.

Independent assessment and fresh retry

W4 requires both inclusive bounds, the available flag, all four outcomes and a distinguishing case with reasoning. After support, retry a 2-to-4-minute inclusive rule with an open flag: (2, True) passes, (4, False) fails and (5, True) fails. Record initial independent work before the retry.

Assess independent transfer

Selected case: Predict all four booleans. Eligibility requires science AND at most 100 pages AND availability. Repeat with genre="history" and all other values unchanged. Which OR result could wrongly suggest the genre requirement passed?

```

pages = 90

genre = "science"

available = False

print(genre == "science" and pages <= 100)

print((genre == "science" and pages <= 100) and available)

print(not available)

print(genre == "science" or pages <= 100)

```

Selected case answer

Original results: True, False, True, True. Full eligibility is False because availability is False. With history: False, False, True, True. The final OR remains True because $90 \leq 100$; it does not establish that the science requirement passed. NOT available is True in both traces.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled boolean conditions evidence sheet to Python readiness portfolio. Show how you can translate stated requirements into comparisons. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use separate cards for each comparison, then combine them. Read at most and at least aloud with number lines. Extension: explain why not $((\text{minutes} \geq 5) \text{ and } (\text{minutes} \leq 9))$ identifies out-of-range numeric values under the stated input contract.

Week 5 uses conditions in if elif and else branches, then tests exact boundaries.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: python-bridge/5, python-bridge/7, python-bridge/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

7. Prepare, teach and assess

Teaching target: Translate stated requirements into comparisons

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

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Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38:

connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time comparison and final suitability values before running the program.

Reasoned practice answer: Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

Check every learner before discussion: Use the worked case for Boolean conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: At most 12 includes 12; compare < and <= at exactly 12.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change age to 16. Require a prediction and a reason before comparing. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.